

Women Safety App - Complete Overview

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1) What this app does

- Provides rapid SOS, multi-channel alerts, live tracking, AI-driven risk checks, and a home-screen panic widget.
- Built with Flutter + Firebase; Android widget via Kotlin + home_widget plugin.

2) Core Features (user-facing)

- SOS: Countdown, loud alarm, multi-channel alerts (WhatsApp/SMS/Email/Push), evidence capture (photo/audio/video), call escalation.
- Location Sharing: One-tap share of current location to guardians with Google Maps link (WhatsApp/SMS).
- Guardians: Add/edit/remove guardians, mark primary, live tracking, route deviation alerts.
- Revolutionary Features Hub (8 items): AI danger prediction, voice distress analysis, face recognition, fake call, safe routes (Google Maps walking directions), evidence capture, live streaming (Agora), volunteer network.
- Ride/Route Safety: Ride tracking, safe journey monitoring, shake-to-SOS, voice activation.
- Panic Home Widget: 2x2 widget with SOS button, status text, contact count; taps trigger panic or launch app.
- Settings & Profile: Theme toggle, permissions, notifications, quick actions, password reset.

3) Key Screens and Routes (MaterialApp routes in lib/app.dart)

- Splash (/), Login (/login), Home (/home), SOS (/sos), Map (/map), Profile (/profile), Settings (/settings), Revolutionary Features (/revolutionary-features), Guardian Management (/guardians), Safety Checklist (/safety-checklist).
- Quick Actions (Profile): Share My Location → /map; Revolutionary Features → /revolutionary-features; Panic Widget Setup dialog.

4) Important Services (lib/services)

- sos_service.dart: Orchestrates 9-step SOS flow, alarm, escalation.
- location_share_service.dart: Builds Google Maps link, shares via WhatsApp/SMS; handles geolocation errors.
- whatsapp_service.dart: Formatted SOS messages, per-contact error handling, delays, success tracking.
- sms_service.dart: Compact SOS text with multiple URI fallbacks (sms:, smsto:, mms:).
- panic_widget_service.dart: Bridges Flutter and Android widget (HomeWidget data storage, trigger checks).
- protection_service.dart: Background/pending SOS trigger checks.
- notification_service.dart: Local/push notification helpers.
- permissions_service.dart: Requests/validates permissions (location, notifications, mic, camera).
- ai_danger_prediction_service.dart / distress_voice_analysis_service.dart / face_recognition_service.dart / live_streaming_service.dart: AI/ML and streaming capabilities (Agora APP_ID required; TFLite model required for danger prediction).
- offline_queue_service.dart: Queues operations when offline.
- guardian_network_service.dart / guardian_tracking_service.dart / ride_tracking_service.dart: Guardian and journey safety.
- fake_call_service.dart: Simulated incoming calls.

5) Android Widget (android/app/src/main)

- PanicWidgetProvider.kt: Handles widget update, click intent (PANIC_BUTTON_CLICKED), SharedPreferences data.
- panic_widget.xml: Layout (red gradient card, circular SOS button, status, contact count).
- panic_widget_info.xml: Metadata (min size, category home_screen, preview).
- Drawables: widget_background.xml (gradient), panic_button_bg.xml (circular red), panic_widget_preview.xml.
- Manifest: Receiver registered with APPWIDGET_UPDATE and custom intent.

6) Setup Requirements

- Firebase: Add android/app/google-services.json (and iOS plist if applicable).
- Permissions: Location, camera, microphone, notifications, vibration; accept runtime prompts.
- Optional: Agora APP_ID in live_streaming_service.dart; TFLite model in assets/models/.

7) How main.dart boots

- Initializes Firebase, notifications, permissions, offline queue, panic widget.
- Checks ProtectionService pending triggers and widget panic triggers.
- Runs MultiBlocProvider (ThemeCubit, AuthCubit, LocationCubit, SosCubit) → WomenSafetyApp.

8) How SOS/Share flows work

- SOS: Trigger → countdown → alarm → evidence capture → multi-channel alerts → escalation calls.
- Share Location: Map screen FAB → loads guardians from Firestore → dialog with WhatsApp/SMS buttons → sends Google Maps link.
- Safe Routes: Revolutionary features FAB → opens Google Maps walking directions from current location.

9) Testing Checklist (manual)

- Profile → Quick Actions → all 3 entries navigate/show dialog.
- Map → Share → choose contact → WhatsApp/SMS opens with link.
- Revolutionary Features → Safe Routes → Google Maps opens.
- Widget → Add to home → tap widget triggers app/intent.
- SOS → triggers alerts (with real credentials and permissions).

10) Known External Requirements

- WhatsApp/SMS require apps installed and phone numbers with country code.
- Firebase init fails without google-services.json.
- Widget visibility depends on launcher supporting home widgets.

File generated for quick reference. For richer visuals, see [project_documentation.html](#) in the same folder.